SOFTWARE REQUIREMENTS SPECIFICATION

CanSupport AI

Affective Computing-Enhanced RAG for Cancer Patient Engagement: An Emotion-Aware Support System for NGOs

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for CanSupport AI, an AI-powered patient engagement platform being developed for Sanjeevani – Life Beyond Cancer, a Mumbai-based NGO supporting cancer patients across India. The document is intended to guide design, development, testing, and academic evaluation of the system over a three-month project timeline.

1.2 Document Conventions

Functional requirements are labelled FR-<module>.<number> (e.g., FR-3.2) and prioritised as High, Medium, or Low. Non-functional requirements are labelled NFR-<number>. "Shall" denotes a mandatory requirement; "should" denotes a recommended one.

1.3 Intended Audience and Reading Suggestions

This document is intended for the student development team and project guide/evaluator who may review functional scope. Sections 1–2 provide context; Section 3 is the core functional specification; Sections 4–8 cover interfaces, quality attributes, data, and the emotional-scoring model; Section 9 is a supporting appendix.

1.4 Project Scope

CanSupport AI will provide a public, login-free patient portal with an AI chatbot capable of answering questions about Sanjeevani's programs, hospitals, and volunteers in English and Hindi. Answers are grounded in Sanjeevani's own verified content using Retrieval-Augmented Generation (RAG), preventing fabricated information. The system additionally analyses the emotional tone of patient messages to adapt its responses and to escalate conversations showing signs of significant distress to a human counsellor. An admin dashboard gives Sanjeevani staff visibility into query volume, response performance, and aggregate emotional trends. The system does not replace counsellors, does not provide medical diagnosis or treatment advice, and does not perform any clinical psychological assessment.

1.5 Definitions, Acronyms and Abbreviations

- RAG — Retrieval-Augmented Generation: a technique that grounds LLM output in retrieved source documents.
- NLP — Natural Language Processing.
- LLM — Large Language Model.
- EDS — Emotional Distress Score, an experimental support-prioritisation metric defined in Section 8.
- SLA — Service Level Agreement (here, an internal response-time target).
- Knowledge Base — the indexed corpus of Sanjeevani's website, program documents, annual reports, and hospital/volunteer data.

1.6 References

- CanSupport AI Project Synopsis (internal document).
- Sanjeevani – Life Beyond Cancer, organisational program materials and annual reports.
- Digital Personal Data Protection Act, 2023 (India).

2. Overall Description

2.1 Product Perspective

CanSupport AI is a new, standalone web platform. It is not a replacement for Sanjeevani's counselling staff but a self-service and triage layer in front of them: it answers what it can verify, and routes the rest — along with an AI-drafted reply and, where relevant, an emotional-urgency flag — to a human.

2.2 Product Functions

- Browse Sanjeevani programs and find nearby volunteers/counsellors without registering.
- Ask free-text questions in English or Hindi and receive grounded, verified answers within seconds.
- Have the system recognise emotional signals (fear, confusion, distress, etc.) and adapt tone and content accordingly.
- Automatically escalate conversations showing sustained or severe distress to a counsellor.
- Give Sanjeevani staff a dashboard of query volume, response times, and aggregate emotional trends.

2.3 User Classes and Characteristics

User Class	Description	Characteristics
Patient / Caregiver	Individual diagnosed with cancer, or a family member/caregiver, seeking program information, hospital contacts, or emotional support.	No login required; varies widely in digital and health literacy; may write in English, Hindi, or Hinglish.
Sanjeevani Counsellor	Trained staff member who receives escalated queries and distress alerts and responds to patients directly.	Uses the admin dashboard daily; needs fast triage and low-friction reply tools.
Programme Supervisor	Oversees multiple counsellors and hospital centres.	Needs queue-wide visibility, analytics, and workload distribution tools.
System Administrator	Manages the knowledge base, user roles, and system configuration.	Technical user responsible for content accuracy and platform uptime.

2.4 Operating Environment

Patient-facing portal: modern mobile and desktop browsers (Chrome, Safari, Edge, Firefox — last two major versions), responsive from 320px width upward. Backend: cloud-hosted web application and API layer, vector database for the knowledge base, and an LLM accessed via API. Admin dashboard: desktop-optimised browser application with role-based login.

2.5 Design and Implementation Constraints

- All chatbot answers must be grounded in the Sanjeevani knowledge base; the system shall not answer from unconstrained LLM general knowledge (see FR-3.3).
- The platform must support English and Hindi at minimum, including code-mixed (Hinglish) input.
- Emotion and distress detection shall be presented strictly as support/prioritisation signals, never as a clinical diagnosis (see FR-4.6).
- The system must be deliverable within a three-month academic project timeline.

2.6 User Documentation

A brief in-portal help/FAQ section for patients, and a counsellor/admin user guide covering the escalation queue, reply workflow, and dashboard, will be delivered alongside the system.

2.7 Assumptions and Dependencies

- Sanjeevani will provide access to program documents, annual reports, and current volunteer-hospital data for the knowledge base.
- An LLM API (or comparable hosted model) and a vector database are available for use during development.
- Counsellors have basic computer literacy sufficient to use a web dashboard.
- Internet connectivity is assumed on the patient's device; the system does not address fully offline use.

3. System Features (Functional Requirements)

Requirements are grouped by the five functional modules of the system.

3.1 Module 1 — Patient Portal

ID	Requirement	Description	Priority
FR-1.1	Program Explorer	Patients can browse all 14 Sanjeevani programs with descriptions, eligibility criteria, and enrolment steps, without login.	High
FR-1.2	Hospital & Volunteer Directory	Patients can search by city or hospital name to find active Sanjeevani volunteers/counsellors, contact details, and visiting schedule.	High
FR-1.3	Cancer Education Resources	Static and curated educational content on cancer types, treatment stages, and survivorship, sourced from verified Sanjeevani material.	Medium

ID	Requirement	Description	Priority
FR-1.4	Embedded AI Chatbot	A persistent chat widget accessible from every portal page, supporting free-text queries in English and Hindi.	High
FR-1.5	Mobile Responsiveness	The portal renders correctly on screen widths from 320px (mobile) to desktop, with no login/registration barrier.	High
FR-1.6	Accessibility Basics	Support for readable font scaling and screen-reader-compatible markup for low-literacy and elderly users.	Medium

3.2 Module 2 — NLP Query Understanding

ID	Requirement	Description	Priority
FR-2.1	Intent Classification	Classify each incoming message into one of: program information, hospital lookup, enrolment request, or emotional support request.	High
FR-2.2	Entity Extraction	Extract key entities such as hospital name, city, cancer type, and program name mentioned in the query.	High
FR-2.3	Language Detection	Automatically detect whether the query is in English, Hindi, or Hinglish (code-mixed), and route the response in the same language.	High
FR-2.4	Urgency/Distress Flagging	Flag messages containing language suggestive of urgency or distress for priority handling by downstream modules.	High
FR-2.5	Query Normalisation	Normalise spelling variants, transliterations, and informal phrasing before passing the query to retrieval.	Medium

3.3 Module 3 — RAG-Based Answer Generation

ID	Requirement	Description	Priority
FR-3.1	Knowledge Base Ingestion	Ingest and index Sanjeevani's website content, program documents, annual reports, and volunteer-hospital data into a searchable vector store.	High
FR-3.2	Semantic Retrieval	Retrieve the most relevant knowledge-base passages for a given query using embedding-based similarity search.	High
FR-3.3	Grounded Answer Generation	Generate a response using an LLM constrained to the retrieved passages only; the system must not answer from unverified general knowledge.	High

ID	Requirement	Description	Priority
FR-3.4	Confidence Scoring	Assign a confidence score to each generated answer based on retrieval relevance; low-confidence answers trigger escalation instead of a guess.	High
FR-3.5	Source Traceability	Every generated answer must be traceable to the specific knowledge-base document(s) used, for audit and correction purposes.	Medium
FR-3.6	Knowledge Base Refresh	Administrators can add, update, or remove source documents; the index refreshes without retraining any model.	Medium

3.4 Module 4 — Affective Computing and Emotional Support

This module is the platform's key differentiator. Rather than treating every query identically, it detects emotional signals in patient messages, tracks them across a conversation, and uses them to adapt how — not just what — the system answers, while ensuring genuinely distressed patients reach a human quickly.

ID	Requirement	Description	Priority
FR-4.1	Emotion Signal Detection	Classify each patient message for emotional signals: fear, sadness, confusion, frustration, hope, uncertainty, and urgency/distress.	High
FR-4.2	Emotion-Adaptive Response Strategy	Use the detected emotion, together with the retrieved knowledge, to select a response style (e.g., simplified reassuring language plus counselling resources vs. a direct factual answer).	High
FR-4.3	Conversation-Level Distress Tracking	Track emotional signals across the turns of a single conversation to detect escalating or sustained distress, not just single-message sentiment.	High
FR-4.4	Emotional Distress Score (EDS)	Compute a running EDS per conversation from emotion intensity, negative sentiment, repeated distress signals, and escalation phrases (see Section 8).	High
FR-4.5	Automatic Human Escalation	When EDS crosses the high-distress threshold, immediately flag the conversation for priority counsellor review via the admin dashboard.	High
FR-4.6	Non-Diagnostic Labelling	All emotion and EDS outputs must be labelled as support/prioritisation signals only, and must never be presented as a clinical or psychiatric diagnosis.	High
FR-4.7	Longitudinal Emotional Profile (Anonymised)	Maintain an anonymised, aggregate view of emotional trends across a patient's journey stages (diagnosis, treatment, post-treatment) for reporting purposes only.	Medium

3.5 Module 5 — Counsellor / Admin Dashboard

ID	Requirement	Description	Priority
FR-5.1	Escalation Queue	Display all chatbot-escalated and distress-escalated conversations in a prioritised queue, ordered by urgency and EDS.	High
FR-5.2	AI-Drafted Reply Suggestion	For each escalated query, present a counsellor with an AI-drafted reply, editable before sending in one click.	High
FR-5.3	Analytics Dashboard	Show query volume by program, hospital, city, language, and time period, plus average and median response time.	High
FR-5.4	Emotional Trend Reporting	Show aggregate, anonymised emotional-signal trends (e.g., anxiety spikes around chemotherapy-stage queries) to management.	Medium
FR-5.5	Role-Based Access Control	Restrict dashboard access by role: counsellor (own queue), supervisor (all queues + analytics), admin (full configuration).	High
FR-5.6	Audit Log	Log every AI-generated answer, escalation, and counsellor action with a timestamp for accountability and quality review.	Medium

4. External Interface Requirements

4.1 User Interfaces

- Patient Portal: responsive React web application; persistent chat widget; language toggle (English/Hindi).
- Admin Dashboard: desktop web application with escalation queue, analytics views, and role-based screens.

4.2 Hardware Interfaces

None beyond standard client devices (smartphone, tablet, desktop/laptop) with a modern web browser and internet access. No specialised hardware is required.

4.3 Software Interfaces

- LLM API for grounded answer generation (Module 3) and emotion-adaptive response drafting (Module 4).
- Vector database / search index for knowledge-base retrieval.
- NLP/emotion-classification service (may be a hosted API or a fine-tuned in-house model).

- Translation/language-detection service for English–Hindi support.

4.4 Communication Interfaces

All client–server communication shall use HTTPS. API calls between internal services (NLP, retrieval, LLM, dashboard) use authenticated REST or equivalent internal protocols.

5. System Architecture Overview

The end-to-end processing pipeline for a single patient message is as follows:

- 1. Patient submits a message via the portal chat widget.
- 2. NLP layer performs language detection, intent classification, and entity extraction (Module 2).
- 3. Affective computing layer analyses emotional signals and updates the conversation's running Emotional Distress Score (Module 4).
- 4. The query, together with detected intent and emotional context, is used to retrieve relevant passages from the knowledge base (Module 3).
- 5. The LLM generates a response grounded in the retrieved passages, adapted in tone and content to the detected emotional state.
- 6. If retrieval confidence is low, or EDS crosses a distress threshold, the conversation is routed to the counsellor escalation queue with an AI-drafted reply (Module 5).
- 7. The response is returned to the patient; all steps are logged for the audit trail and analytics.

6. Non-Functional Requirements

6.1 Performance

- NFR-1: The chatbot shall return a response within 5 seconds under normal load for at least 95% of queries.
- NFR-2: The platform shall support at least 200 concurrent chat sessions without degradation, scalable via cloud infrastructure.

6.2 Security and Privacy

- NFR-3: All data in transit shall be encrypted via HTTPS/TLS; sensitive data at rest shall be encrypted.
- NFR-4: The patient portal shall not require login or collect identifying information beyond what a patient voluntarily provides in a message.
- NFR-5: Dashboard access shall be role-based and authenticated (see FR-5.5).
- NFR-6: Personal and health-related information shall be handled in line with India's Digital Personal Data Protection Act, 2023.

6.3 Usability and Accessibility

- NFR-7: The portal shall be usable by patients with limited digital literacy, using plain-language UI copy in both English and Hindi.

- NFR-8: Text shall meet basic accessibility contrast and scaling standards for elderly and low-vision users.

6.4 Reliability and Availability

- NFR-9: The platform shall target 99% uptime during business hours.
- NFR-10: If the LLM or retrieval service is unavailable, the chatbot shall degrade gracefully to a static FAQ and offer a direct escalation option rather than failing silently.

6.5 Scalability

- NFR-11: The knowledge base shall be updatable (add/edit/remove documents) without retraining any model, per FR-3.6.

6.6 Maintainability

- NFR-12: Modules 2–5 shall be loosely coupled services so that the NLP, retrieval, affective-computing, or dashboard components can be updated independently.

6.7 Ethical AI and Safety Requirements

- NFR-13: The system shall never present emotion detection or the EDS as a clinical or psychiatric diagnosis, per FR-4.6.
- NFR-14: The system shall never fabricate program eligibility, medical, or financial-assistance information not present in the verified knowledge base.
- NFR-15: Conversations flagged as high-distress (EDS 81–100) shall be surfaced to a human counsellor as a priority action, not resolved by AI alone.
- NFR-16: All AI-generated content directed at patients shall be reviewable and correctable by Sanjeevani staff.

7. Data Requirements

7.1 Knowledge Base Data

Sourced from Sanjeevani's website, the 14 program documents, annual reports, and volunteer-hospital directory data. Each source document is indexed with metadata (program, hospital, region, last-updated date) to support traceability (FR-3.5) and targeted refresh (FR-3.6).

7.2 Patient Interaction Data

Conversation transcripts, detected intent, detected language, emotion labels, and EDS values are stored per conversation to support escalation, analytics, and quality review. Conversations are not linked to patient identity unless the patient voluntarily provides contact details for follow-up.

7.3 Data Retention and Privacy

Interaction data used for aggregate analytics and emotional-trend reporting (FR-4.7, FR-5.4) shall be anonymised before being surfaced on the management dashboard. Retention periods and deletion procedures for raw conversation logs shall follow Sanjeevani's data policy and applicable Indian data-protection law.

8. Emotional Distress Score (EDS) Specification

The EDS is an experimental, non-clinical metric used solely to help prioritise counsellor attention. It is computed per conversation from four weighted inputs: emotion intensity, negative sentiment, repeated distress signals across turns, and the presence of explicit escalation phrases. It is reported to counsellors as a prioritisation aid only and must always be labelled as such (FR-4.6, NFR-13).

EDS Range	Category	System Action
0 – 30	Normal conversation	No action; standard chatbot flow continues.
31 – 60	Emotional support recommended	Chatbot offers counselling-resource information alongside its answer.
61 – 80	Counsellor attention recommended	Conversation is added to the escalation queue for review within the standard SLA.
81 – 100	Priority human review	Conversation is flagged for immediate counsellor attention; AI continues to respond supportively while awaiting human contact.

9. Appendix

9.1 Use Case Summary

- UC-1: A newly diagnosed patient asks a general "what do I do first" question and receives programme guidance plus nearby volunteer contacts.
- UC-2: A caregiver asks about financial assistance for chemotherapy and receives grounded eligibility information or escalation if the knowledge base is inconclusive.
- UC-3: A patient expresses fear the night before chemotherapy; the system responds with simplified, reassuring information and offers counselling resources.
- UC-4: A patient's messages show escalating distress across a conversation; the system raises EDS, continues to respond supportively, and immediately flags the conversation for priority counsellor review.
- UC-5: Sanjeevani management reviews the dashboard to see which programs and hospitals generate the most queries, and which stages of the patient journey show elevated anxiety.

9.2 Assumptions Log

Recorded assumptions (availability of Sanjeevani content, LLM/vector-database access, counsellor digital literacy, and internet-connected patient devices) are listed in Section 2.7 and will be revisited if project constraints change.

9.3 Glossary

See Section 1.5 for core definitions and acronyms used throughout this document.